Ford GoBike Exploratory Data Analysis (EDA) Using Python

Comprehensive Business Intelligence & Data Analytics Report

Tools Used

Python
Pandas
NumPy
Matplotlib
Seaborn
Jupyter Notebook

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1. Introduction

Urban mobility has undergone a significant transformation over the past decade, with bike-sharing programs emerging as a practical solution to congestion, pollution, and the demand for flexible short-distance travel. Ford GoBike, one of the pioneering bike-share systems on the United States West Coast, allows residents and visitors to rent bicycles from a dense network of docking stations spread across San Francisco and neighbouring cities. As ridership has grown, so too has the volume of operational data generated with every trip, creating an opportunity for organizations to apply analytical techniques to better understand how the service is actually used.

Exploratory Data Analysis (EDA) forms the backbone of any sound analytics workflow. Rather than jumping directly into modelling, EDA emphasises careful examination of the data's structure, quality, and underlying patterns. This includes assessing missing values, identifying outliers, understanding distributions, and visualising relationships between variables. For an operational business such as Ford GoBike, EDA is not merely an academic exercise; it directly informs decisions about where bikes should be stationed, when additional staff are required for rebalancing, and which customer segments should be prioritised for retention campaigns.

Business analytics, more broadly, is concerned with converting data into insight and insight into action. This project sits squarely within that discipline: it moves from raw, disparate monthly trip files to a single, cleaned dataset, and then to a structured set of visual analyses that expose meaningful patterns in ridership. The project matters because it demonstrates, end to end, how a real-world transportation dataset can be transformed into a foundation for evidence-based decision-making — a capability that is increasingly essential for any modern, data-driven organisation operating in the shared-mobility space.

2. Problem Statement

Ford GoBike generates substantial volumes of trip-level data every single day, and embedded within these records is valuable information about customer behaviour, ride patterns, station utilisation, and overall service demand. However, raw transactional data on its own offers little direct value; without structured analysis, patterns such as peak commuting windows, station congestion, or shifts in customer composition remain invisible to decision-makers.

The central challenge addressed in this project is to convert a year's worth of fragmented monthly trip records into a unified, analysis-ready dataset, and to then apply Exploratory Data Analysis techniques to surface trends in ride frequency, duration, station popularity, and rider demographics. By resolving data quality issues and systematically visualising the cleaned data, this project aims to equip Ford GoBike's operations and strategy teams with a clear, evidence-based understanding of how the service is used, enabling more informed decisions around resource allocation, fleet rebalancing, and customer engagement.

3. Business Objectives

The following objectives guided the analytical scope and direction of this project:

1. To analyse the demographic profile of riders, including user type, age, and gender.
2. To identify ride patterns across different months, weekdays, and hours of the day.
3. To examine ride duration and understand how it varies among different customer groups.
4. To identify the most frequently used start and end stations for efficient resource planning.
5. To explore relationships between different variables and uncover meaningful trends.
6. To generate actionable business insights that improve operational efficiency and customer experience.
7. To provide data-driven recommendations that support better decision-making and optimise the Ford GoBike bike-sharing service.

4. Dataset Description

The dataset used in this project was assembled from twelve separate monthly CSV files, each representing one calendar month of Ford GoBike trip activity. These files were programmatically located and merged into a single unified DataFrame using Python's glob and pandas libraries, producing a combined dataset that spans a full year of operations and contains well over two million individual trip records. Each record corresponds to one completed bicycle rental and captures a rich set of attributes describing that trip.

Key variables include the trip duration in seconds, the start and end timestamps, the identifiers, names, and geographic coordinates of the start and end stations, the unique bike identifier used for the trip, the user type (distinguishing between registered Subscribers and casual Customers), and rider demographic fields such as gender and birth year. Beyond these sixteen or more raw attributes, several derived features were later engineered to support temporal and demographic analysis. The scale and granularity of this dataset make it well suited to uncovering both broad seasonal patterns and fine-grained behavioural differences between customer segments, forming a solid foundation for the exploratory analysis that follows.

Attribute	Description
Dataset	Ford GoBike Trip Data
Records	2+ Million
Variables	16+ (raw and engineered)
Time Period	One Full Year
Source	Ford GoBike (monthly CSV exports)

Table 4.1: Summary of Dataset Attributes

5. Tools & Technologies

This project was executed entirely within the Python data-science ecosystem, chosen for its maturity, extensive library support, and strong community adoption in analytics workflows. Each tool was selected for a specific role in the pipeline, from raw data ingestion through to the production of polished statistical graphics.

Tool	Purpose
Python	Core programming language for the analysis
Pandas	Data manipulation, merging, and cleaning
NumPy	Numerical computation and array operations

Tool	Purpose
Matplotlib	Foundational data visualisation
Seaborn	Statistical and aesthetic visualisation
Jupyter Notebook	Interactive development and documentation

Table 5.1: Tools and Technologies Used

6. Project Workflow

The project followed a structured, sequential workflow that mirrors industry best practice for exploratory data analysis. Each stage builds on the output of the previous one, ensuring that insights generated later in the process rest on a foundation of clean, well-understood data.

Load Data → Merge → Clean → Feature Engineering → EDA → Business Insights → Recommendations

Load Data

Twelve monthly CSV files were located programmatically and read into individual pandas DataFrames, establishing the raw input for the pipeline.

Merge

The individual monthly DataFrames were concatenated into a single unified dataset, producing a continuous, year-long view of trip activity rather than twelve disconnected snapshots.

Clean

Missing values, duplicate records, invalid ages, and inconsistent data types were systematically identified and resolved to ensure the dataset was reliable for analysis.

Feature Engineering

New analytical fields — including rider age, ride duration in minutes, and time-based groupings such as hour, day, and month — were derived from the existing raw fields.

EDA

Univariate, bivariate, and multivariate visualisations were produced to examine distributions, relationships, and correlations across the cleaned and enriched dataset.

Business Insights

Patterns observed during the visual analysis were synthesised into concise, decision-relevant findings for Ford GoBike's operational and strategic teams.

Recommendations

Finally, the insights were translated into specific, actionable recommendations designed to improve fleet management, customer engagement, and long-term planning.

7. Data Collection & Integration

The raw data for this project arrived as twelve separate monthly extracts stored in comma-separated value (CSV) format, each representing one month of Ford GoBike operations. Rather than analysing these files in isolation, the first technical task was to build a repeatable integration process capable of combining them into a single, analysis-ready structure. Python's built-in glob module was used to programmatically discover every CSV file within the designated data directory, removing the need to reference file names manually and making the pipeline resilient to the addition of future monthly extracts.

Each discovered file was read into a separate pandas DataFrame and stored in a list, after which the pandas concat function was used to stack all twelve DataFrames into one continuous dataset with a reset, contiguous index. This step was essential because bike-share usage is inherently seasonal, and any analysis limited to a single month would misrepresent the broader annual pattern. Following the merge, an initial inspection of the dataset's shape confirmed that the integration had produced a dataset with well over two million rows, matching expectations for a full year of urban bike-share activity. This integrated dataset then served as the single source of truth for all subsequent cleaning, feature engineering, and exploratory analysis performed in this project.

8. Data Cleaning

With the twelve monthly files successfully merged, attention turned to assessing and improving data quality. An initial audit of the combined dataset revealed missing values concentrated primarily in the member_gender and member_birth_year fields, along with a small number of duplicate trip records introduced during the merge process. Left unaddressed, these issues would have distorted demographic analysis and inflated ride counts, so a systematic cleaning routine was applied before any visualisation work began.

Rows missing gender or birth-year information were removed using pandas' dropna function, since these fields are central to the demographic analysis at the heart of this project and could not be reliably imputed without introducing bias. Duplicate rows were identified with the duplicated method and eliminated using drop_duplicates, ensuring that each trip was represented exactly once. The start_time and end_time columns, originally stored as plain text, were converted to proper datetime objects using pandas' to_datetime function, unlocking the ability to extract hour, day, and month values later in the pipeline. Rider age was then derived from birth year, and implausible values — ages below ten or above one hundred — were filtered out as likely data-entry errors. Finally, any trips with a non-positive recorded duration were removed, since these represent logging anomalies rather than genuine rides. Together, these steps produced a dataset that was consistent, well-typed, and trustworthy for the analysis that followed.

Issue	Solution
Missing Values	Rows with missing gender or birth year handled via removal
Duplicates	Duplicate trip records identified and removed
Invalid Ages	Records outside a plausible 10–100 age range filtered out
Datatypes	Timestamp fields converted to proper datetime objects
Date Columns	start_time and end_time parsed for temporal feature extraction

Table 8.1: Data Cleaning Summary

9. Feature Engineering

Once the dataset had been cleaned, several new variables were engineered to support richer temporal and demographic analysis. The raw duration_sec field, while precise, is not intuitive for interpretation, so it was converted into ride_duration_min by dividing by sixty, giving a measure of trip length that is far easier to reason about in minutes rather than seconds.

Temporal features were extracted directly from the parsed start_time timestamp. The ride_hour field captures the hour of day at which each trip began, enabling analysis of intraday demand cycles such as morning and evening commuting peaks. The ride_day field records the day of the week, supporting comparisons between weekday and weekend usage, while ride_month captures the calendar month, allowing seasonal ridership trends to be examined across the full year of data. In parallel, rider age was calculated by subtracting each member's birth year from the reference year of the dataset, producing a continuous demographic variable suitable for both distributional analysis and correlation studies. Collectively, these engineered features transformed a purely transactional dataset into one capable of answering meaningful business questions about when, how long, and by whom the Ford GoBike service is used.

Feature	Purpose
ride_month	Supports monthly and seasonal ridership analysis
ride_day	Supports weekly / weekday-versus-weekend analysis
ride_hour	Supports hourly and peak-demand trend analysis
age	Supports demographic and age-based analysis
ride_duration_min	Supports ride duration and usage-behaviour analysis

Table 9.1: Engineered Features

10. Exploratory Data Analysis

This section presents the complete set of visual analyses performed on the cleaned Ford GoBike dataset. The analysis progresses from univariate examination of individual variables, through bivariate comparisons between pairs of variables, to multivariate exploration of relationships across several variables simultaneously. Each visualisation is accompanied by a summary of key statistics, a detailed interpretation, a set of concise observations, and a corresponding business recommendation.

10.1 Univariate Analysis

10.1 Distribution of User Types

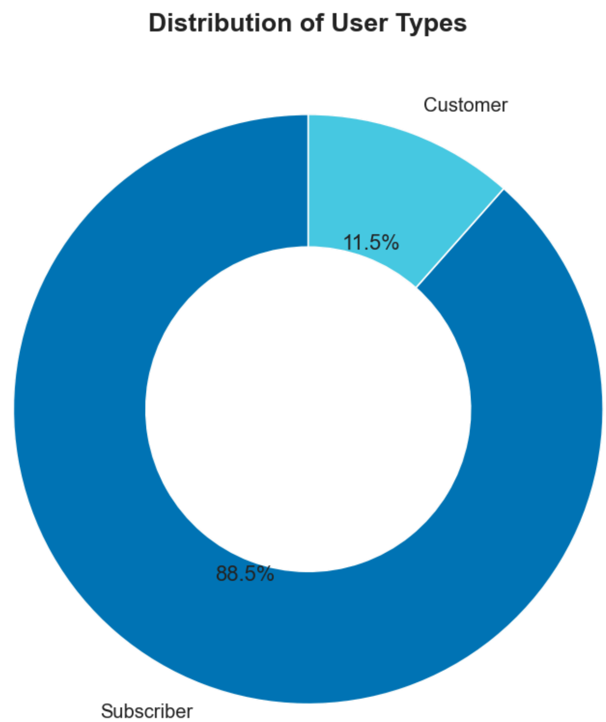

Figure 10.1: Distribution of User Types

Metric	Observation
Subscriber Share	88.5% of all recorded trips
Customer Share	11.5% of all recorded trips
Dominant Segment	Subscriber

Table 10.1: Summary Statistics — Distribution of User Types

Interpretation

This donut chart presents the proportional split between the two categories of Ford GoBike users: Subscribers, who hold ongoing membership plans, and Customers, who pay for individual or short-term access. The visualisation shows an overwhelming concentration of ridership among Subscribers, who account for 88.5 percent of all trips, while Customers contribute the remaining 11.5 percent. This imbalance indicates that the service's core usage base is built around repeat, habitual riders rather than one-off or tourist-style users. From a business perspective, this pattern is significant because subscription-based riders generally represent a more predictable and recurring source of revenue than pay-per-ride customers. It also suggests that Ford GoBike has succeeded in embedding itself into the daily routines of a large segment of its user base, likely for commuting or regular short-distance travel. At the same time, the relatively small proportion of Customers represents an underexploited opportunity: casual riders who have not yet converted to subscribers could be targeted with tailored incentives. Understanding this split is foundational to the remainder of the analysis, since rider behaviour, trip duration, and station preferences frequently differ meaningfully between these two groups, as later visualisations in this report will demonstrate.

Key Observations

- Subscribers represent the clear majority of total trips at 88.5 percent.
- Casual Customers make up a comparatively small 11.5 percent share.
- The heavy skew toward Subscribers suggests strong daily/commuter reliance on the service.
- Customer segment, while smaller, remains a meaningful pool for conversion campaigns.

Business Recommendation

- Design loyalty incentives and referral discounts to convert frequent Customers into Subscribers.
- Protect subscriber satisfaction through reliable bike availability, since this group anchors overall revenue.

10.2 Distribution of Rider Gender

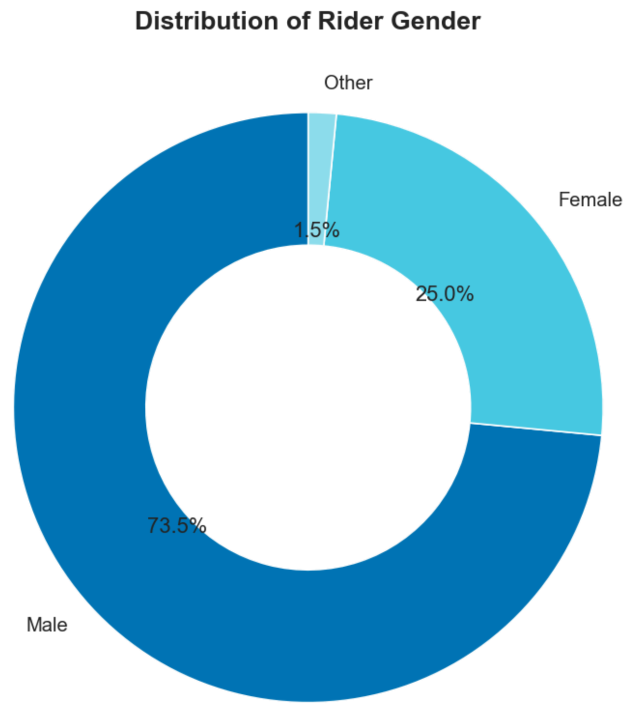

Figure 10.2: Distribution of Rider Gender

Metric	Observation
Male Riders	73.5% of identified riders
Female Riders	25.0% of identified riders
Other	1.5% of identified riders

Table 10.2: Summary Statistics — Distribution of Rider Gender

Interpretation

This chart breaks down the rider base by self-reported gender, distinguishing between Male, Female, and Other categories. The results show a pronounced skew toward male riders, who make up 73.5 percent of the dataset, compared to 25.0 percent for female riders and a small 1.5 percent identifying as Other. Such an imbalance is common across many bike-share systems globally and often reflects a combination of factors, including differences in commuting patterns, comfort levels with urban cycling infrastructure, and historical marketing approaches that may have skewed toward a particular demographic. For Ford GoBike, this distribution highlights a clear opportunity area: efforts to grow ridership among women and non-binary riders could unlock meaningful, currently underserved demand. Understanding this composition is also important for safety and infrastructure planning, as different rider groups may have different preferences regarding station placement, lighting, and route design. While this chart does not by itself explain the underlying causes of the imbalance, it provides a quantitative baseline against which future diversity and inclusion initiatives can be measured, and it should be considered alongside the age and duration analyses presented later in this report.

Key Observations

- Male riders dominate the platform, accounting for nearly three-quarters of trips.
- Female ridership, at one quarter of the base, represents a substantial but secondary segment.
- Riders identifying as Other form a very small proportion of the total.
- The gender skew is consistent with patterns commonly observed in other bike-share systems.

Business Recommendation

- Launch targeted marketing and safety-focused campaigns to grow ridership among underrepresented groups.
- Track gender composition over time to measure the impact of inclusion-focused initiatives.

10.3 Age Distribution of Riders

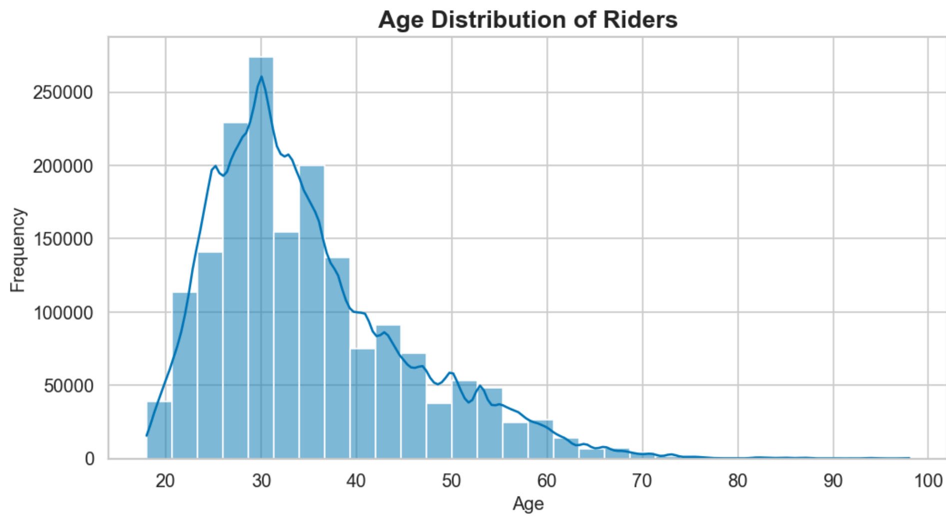

Figure 10.3: Age Distribution of Riders

Metric	Observation
Most Common Age Range	Approximately 25–35 years
Distribution Shape	Right-skewed
Minimum Age (filtered)	10 years

Table 10.3: Summary Statistics — Age Distribution of Riders

Interpretation

The histogram of rider age, overlaid with a smoothed density curve, reveals that the bulk of Ford GoBike's user base falls between the mid-twenties and mid-thirties, with the frequency of riders tapering off steadily as age increases beyond that range. The distribution is clearly right-skewed: while the vast majority of trips are taken by riders under forty, a long tail extends out toward riders in their sixties, seventies, and even beyond ninety, though these older age groups contribute comparatively few trips. This pattern is consistent with the demographic profile typically associated with urban cycling and daily commuting, where younger working-age adults are the most frequent participants. The concentration of ridership in this age band has direct implications for how Ford GoBike designs its marketing messaging, mobile app experience, and even physical bike design, since the preferences of a 28-year-old commuter may differ substantially from those of a 60-year-old recreational rider. The presence of a long tail of older riders, while numerically small, also suggests that the service has some appeal across a broader age spectrum and should not be entirely overlooked when planning inclusive outreach programmes.

Key Observations

- The 25–35 age band represents the single largest concentration of riders.
- Ridership declines steadily and consistently after age 40.
- A long but thin tail of older riders extends into the 60–90+ range.
- The overall shape confirms a young, working-age core user base.

Business Recommendation

- Prioritise marketing channels and messaging that resonate with young working professionals.
- Explore lightweight outreach for older riders to broaden the demographic base over time.

10.4 Monthly Ride Distribution

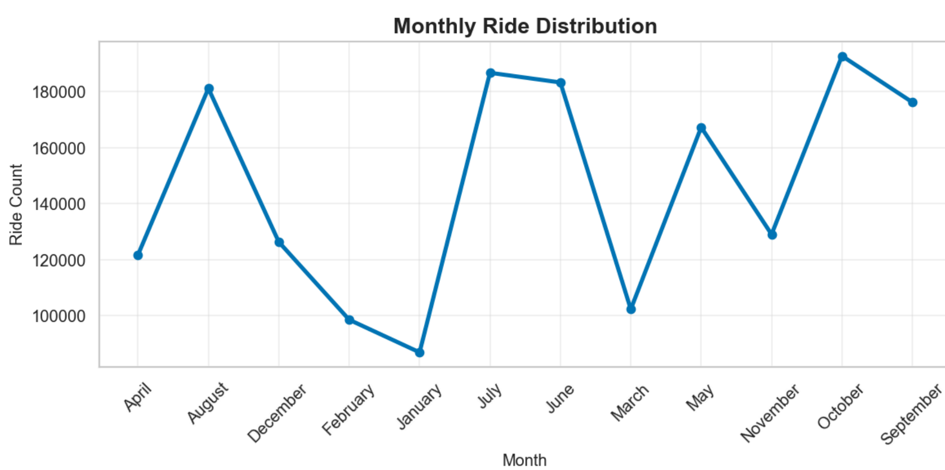

Figure 10.4: Monthly Ride Distribution

Metric	Observation
Peak Month	October (highest ride count)
Lowest Month	January (lowest ride count)

Metric	Observation
Overall Pattern	Seasonal, with a spring–autumn peak

Table 10.4: Summary Statistics — Monthly Ride Distribution

Interpretation

This line chart tracks total ride volume across each calendar month, revealing a distinctly seasonal pattern in Ford GoBike usage. Ridership is at its lowest in January and February, climbs sharply into the spring, and reaches its highest points in the mid-to-late part of the year, with October recording the largest overall ride count, closely followed by July and June. A brief but notable dip occurs in March before demand rebounds again heading into summer. This seasonal rhythm is likely driven by weather conditions, since cooler, wetter months in the Bay Area tend to discourage casual and commuter cycling alike, while milder conditions in the warmer months make cycling a more attractive transportation option. From a planning standpoint, this pattern is highly actionable: it implies that Ford GoBike should scale its operational capacity, including bike availability, maintenance staffing, and station servicing, upward heading into the spring and summer months, and can afford to scale back somewhat during the winter trough. Recognising this cyclical demand also supports better financial forecasting, since revenue from ride fees and subscription usage will naturally ebb and flow in line with these seasonal patterns.

Key Observations

- Ridership is lowest in the January–February winter period.
- Demand rises sharply through spring and peaks around October.
- A temporary dip appears in March before the summer rebound.
- The overall pattern strongly suggests a weather-driven seasonal cycle.

Business Recommendation

- Scale fleet size and station servicing upward ahead of the spring and summer demand surge.
- Use quieter winter months for fleet maintenance, station upgrades, and system improvements.

10.2 Bivariate Analysis

10.5 Weekday Ride Distribution

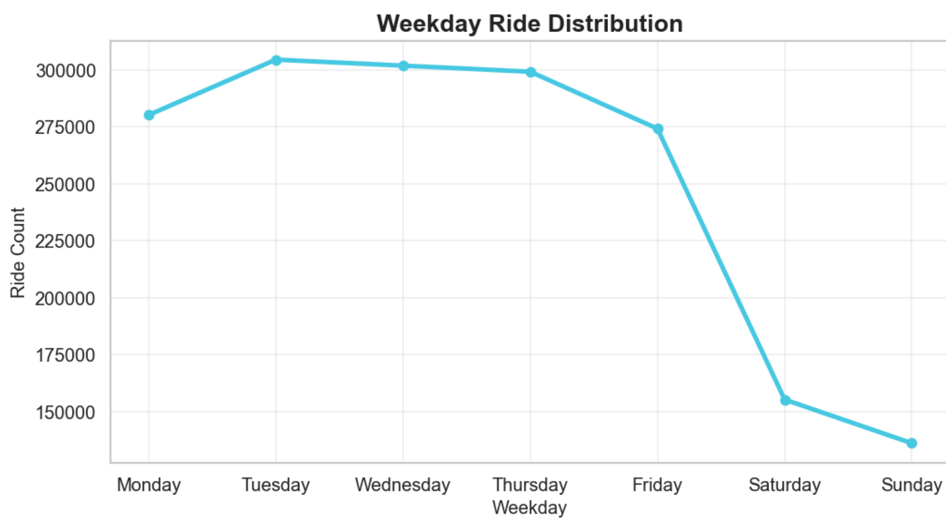

Figure 10.5: Weekday Ride Distribution

Metric	Observation
Peak Day	Tuesday
Lowest Day	Sunday
Pattern	Weekday-dominant usage

Table 10.5: Summary Statistics — Weekday Ride Distribution

Interpretation

This chart compares total ride volume across each day of the week and reveals a clear weekday-versus-weekend divide. Ride counts remain consistently high from Monday through Thursday, with Tuesday recording the single highest volume, before declining modestly on Friday and then dropping sharply on Saturday and Sunday. This steep weekend decline, roughly halving the ride count compared to a typical weekday, strongly suggests that the majority of trips are tied to work-related commuting rather than leisure or tourism. Such a finding aligns naturally with the earlier observation that Subscribers dominate the user base, since subscription-based commuters are far more likely to ride on a fixed weekday schedule than casual weekend visitors. This weekday concentration carries important operational implications: bike availability, station capacity, and rebalancing operations need to be most robust from Monday to Friday, while weekend operations can potentially run with a leaner deployment. It also opens a strategic question for Ford GoBike around whether targeted weekend promotions could help even out demand and better utilise the fleet across all seven days rather than allowing bikes to sit idle during the quieter weekend period.

Key Observations

- Weekday ridership (Monday–Thursday) remains consistently high.
- Tuesday emerges as the single busiest day of the week.
- Ridership falls markedly on Saturday and Sunday.
- The pattern reinforces that commuting, not leisure, drives most demand.

Business Recommendation

- Maintain maximum fleet availability and rebalancing coverage on weekdays.
- Introduce weekend promotions or leisure-oriented pricing to boost off-peak utilisation.

10.6 Hourly Ride Distribution

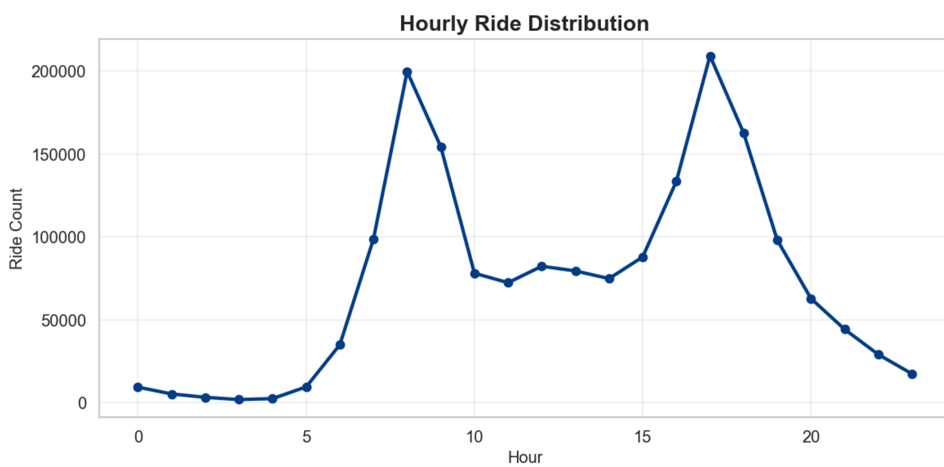

Figure 10.6: Hourly Ride Distribution

Metric	Observation
Morning Peak	Around 8:00 AM

Metric	Observation
Evening Peak	Around 5:00 PM
Quietest Hours	Midnight to 4:00 AM

Table 10.6: Summary Statistics — Hourly Ride Distribution

Interpretation

The hourly ride distribution chart uncovers a clear bimodal pattern, with two sharply defined peaks: one in the early morning around eight o'clock and a second, slightly larger peak in the late afternoon around five o'clock. Between these two peaks, ridership dips to a moderate plateau through the midday hours, before falling away almost to zero overnight between midnight and roughly four in the morning. This double-peaked shape is the classic signature of a commuter-driven transportation system, with riders travelling to work or school in the morning and returning home in the early evening. The near-total absence of activity overnight confirms that Ford GoBike is used almost exclusively during waking, active hours rather than as a round-the-clock mobility option. These findings are highly relevant for operational planning: bike redistribution efforts should be timed to ensure stations are well stocked immediately before the morning peak and that docking capacity is available before the evening peak, when returning riders will be concentrated at particular destinations. Staffing for maintenance and rebalancing crews can likewise be concentrated around these two windows rather than spread evenly across a 24-hour cycle.

Key Observations

- Two distinct peaks occur at approximately 8 AM and 5 PM.
- The evening peak is marginally higher than the morning peak.
- Overnight hours see minimal to negligible ridership.
- The bimodal shape is characteristic of commuter travel behaviour.

Business Recommendation

- Schedule bike redistribution to be complete just before the 8 AM and 5 PM demand windows.

- Concentrate maintenance and rebalancing staff shifts around these two peak periods.

10.7 Ride Duration Distribution

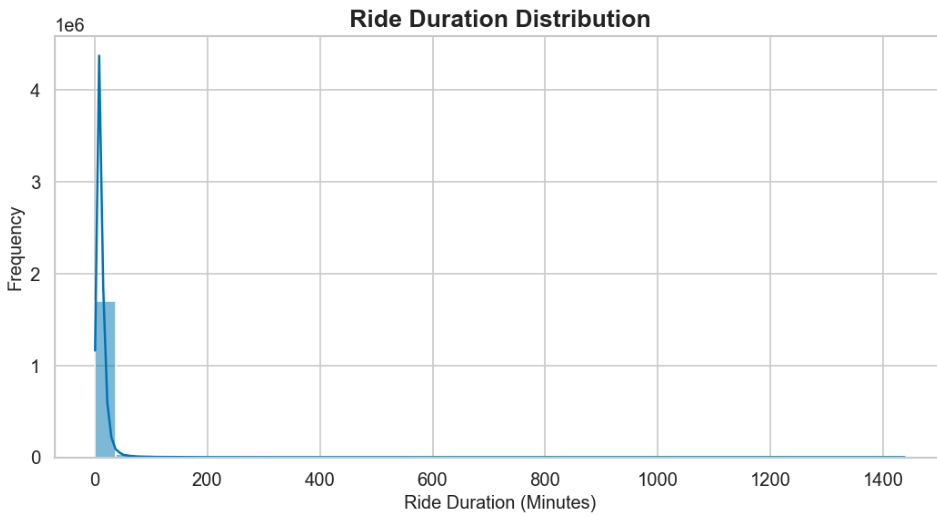

Figure 10.7: Ride Duration Distribution

Metric	Observation
Most Common Duration	Under 15 minutes
Distribution Shape	Heavily right-skewed
Long-duration Trips	Rare but present in the tail

Table 10.7: Summary Statistics — Ride Duration Distribution

Interpretation

This histogram, overlaid with a density curve, displays the distribution of ride duration across all trips. The overwhelming majority of rides are extremely short, clustering well under fifteen minutes, after which the frequency of longer trips falls away rapidly. The distribution is heavily right-skewed, with a long but very thin tail extending out toward several hundred minutes, indicating that while most trips are brief, a small number of outlier rides last considerably longer. This shape is entirely consistent with the system's evident role as a short-distance, point-to-point transportation tool rather than a leisurely, extended-use rental service. The dominance of brief trips supports the earlier hypothesis that most riders use Ford GoBike for practical commuting or last-mile connectivity, quickly checking out a bike, completing a short journey, and returning it promptly, which also has favourable implications for bike turnover and fleet utilisation efficiency. The presence of the long tail, however, warrants closer attention, since these extended rentals may indicate lost or forgotten bikes, unusual usage patterns, or billing anomalies that could affect revenue and fleet availability if left unmonitored.

Key Observations

- The vast majority of trips last under fifteen minutes.
- Frequency declines sharply as duration increases beyond this range.
- A thin but persistent tail of long-duration trips exists in the data.
- The overall shape confirms short-distance, utilitarian usage as the primary use case.

Business Recommendation

- Optimise fleet turnover strategies around the dominant short-trip usage pattern.
- Implement automated alerts for unusually long rentals to catch potential lost-bike incidents early.

10.8 Ride Duration Boxplot

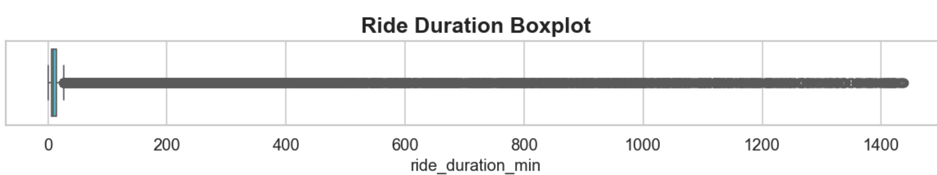

Figure 10.8: Ride Duration Boxplot

Metric	Observation
Median Duration	Low, tightly clustered near the lower end
Interquartile Range	Narrow, concentrated at short durations
Outliers	Numerous, extending beyond 1,000 minutes

Table 10.8: Summary Statistics — Ride Duration Boxplot

Interpretation

This boxplot provides a compact statistical summary of ride duration and, in doing so, starkly illustrates the extent of outlier activity within the dataset. The box itself, representing the interquartile range, sits tightly compressed near the lower end of the duration scale, confirming that the central fifty percent of trips are short in length and closely clustered together. What stands out most dramatically, however, is the dense band of individual outlier points stretching across the chart, with some rides extending beyond 1,000 minutes and approaching the maximum recorded duration of roughly 1,440 minutes, equivalent to a full 24-hour period. This visualisation reinforces the finding from the duration histogram but makes the scale of the outlier problem immediately visible: a non-trivial number of rentals appear to have lasted an entire day or close to it. Such extreme durations are unlikely to reflect genuine continuous cycling and more plausibly indicate bikes that were checked out and not promptly returned, lost, or involved in some form of billing or logging irregularity. Identifying and understanding these outliers is important both for accurate reporting of typical usage and for operational follow-up on the affected bikes and accounts.

Key Observations

- The interquartile range is narrow and concentrated at short durations.
- A substantial number of outlier trips extend well beyond typical usage.
- Some outliers approach the maximum possible duration of 24 hours.
- The scale of outliers suggests operational issues such as unreturned bikes.

Business Recommendation

- Introduce automated flags and grace-period policies for rentals exceeding a defined duration threshold.
- Investigate high-duration outliers individually to recover bikes and correct billing where appropriate.

10.9 Top 10 Start Stations

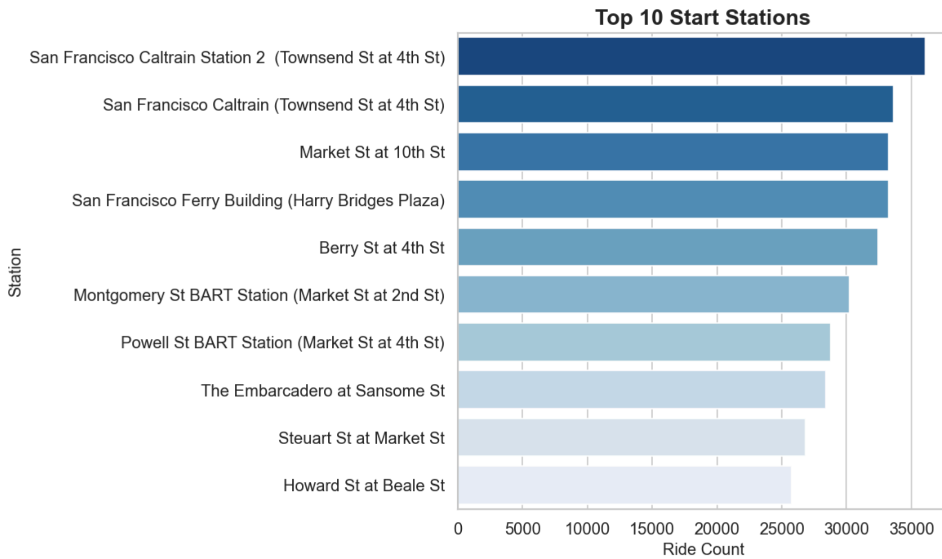

Figure 10.9: Top 10 Start Stations

Metric	Observation
Busiest Start Station	San Francisco Caltrain Station 2 (Townsend St at 4th St)
Common Theme	Transit hubs and downtown commercial areas
Spread	Top ten stations show a gradual, not extreme, decline

Table 10.9: Summary Statistics — Top 10 Start Stations

Interpretation

This horizontal bar chart ranks the ten stations that generated the highest number of trip origins across the year. San Francisco Caltrain Station 2 at Townsend Street and 4th Street leads the list, closely followed by the primary San Francisco Caltrain station and a cluster of well-known downtown locations including Market Street at 10th Street, the San Francisco Ferry Building, and Berry Street at 4th Street. A clear thematic pattern emerges from this ranking: the busiest start stations are overwhelmingly located near major transit interchanges, ferry terminals, and dense commercial districts, reinforcing the earlier finding that Ford GoBike functions primarily as a first-and-last-mile connector for commuters transferring between transit modes. The relatively gradual decline in ride counts moving down the top-ten list, rather than a sharp cliff after the top one or two stations, indicates that demand is spread across several genuinely high-traffic hubs rather than being concentrated at a single dominant location. This has direct implications for fleet distribution: these top stations require consistently high bike availability throughout the day, particularly ahead of the morning commute, and should be prioritised in any rebalancing schedule.

Key Observations

- Transit-adjacent stations near Caltrain and the Ferry Building dominate the top rankings.
- The top ten stations show a gradual rather than a steep decline in ride volume.
- Downtown commercial corridors also feature prominently among top-performing stations.
- The pattern confirms Ford GoBike's role as a transit-connection service.

Business Recommendation

- Prioritise these top stations for morning bike replenishment ahead of the commuter rush.
- Consider expanding dock capacity at consistently high-traffic transit-adjacent stations.

10.10 Top 10 End Stations

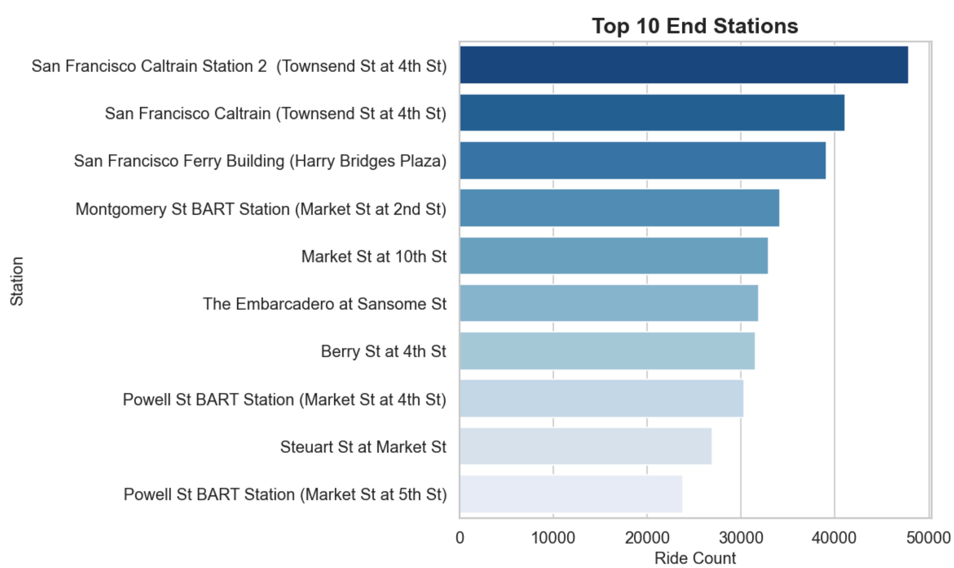

Figure 10.10: Top 10 End Stations

Metric	Observation
Busiest End Station	San Francisco Caltrain Station 2 (Townsend St at 4th St)

Metric	Observation
Common Theme	Same transit hubs dominate as destinations
Notable Shift	Montgomery St BART Station ranks higher as a destination

Table 10.10: Summary Statistics — Top 10 End Stations

Interpretation

This chart mirrors the previous analysis but focuses on the ten stations that most frequently serve as trip destinations rather than origins. Unsurprisingly, San Francisco Caltrain Station 2 again tops the list, confirming its role as the single most critical node in the network, whether riders are beginning or ending their journeys there. The broader list of top end stations closely echoes the top start stations, with the Ferry Building, Market Street at 10th Street, and Berry Street at 4th Street all reappearing, though in a slightly different order. One noteworthy shift is that Montgomery Street BART Station ranks more prominently as a destination than it does as an origin, suggesting that a meaningful volume of riders use Ford GoBike specifically to reach this transit connection point at the end of their trip, likely for onward travel via BART. The strong overlap between top start and end stations indicates that these hubs function as two-way commuter gateways rather than one-directional feeders, which is an important consideration for balancing bike inventory, since bikes arriving at these stations during peak hours must be efficiently redistributed to prevent docks from overflowing or from running empty ahead of the next demand cycle.

Key Observations

- The same station, San Francisco Caltrain Station 2, tops both the start and end rankings.
- Most top destination stations also appear among the top origin stations.
- Montgomery St BART Station shows a relatively stronger role as a destination.
- The overlap points to these hubs functioning as two-directional commuter gateways.

Business Recommendation

- Deploy rebalancing crews proactively at top end stations to prevent dock saturation during peak arrival windows.

- Coordinate station capacity planning jointly across top start and end locations rather than independently.

10.11 User Type vs Ride Duration

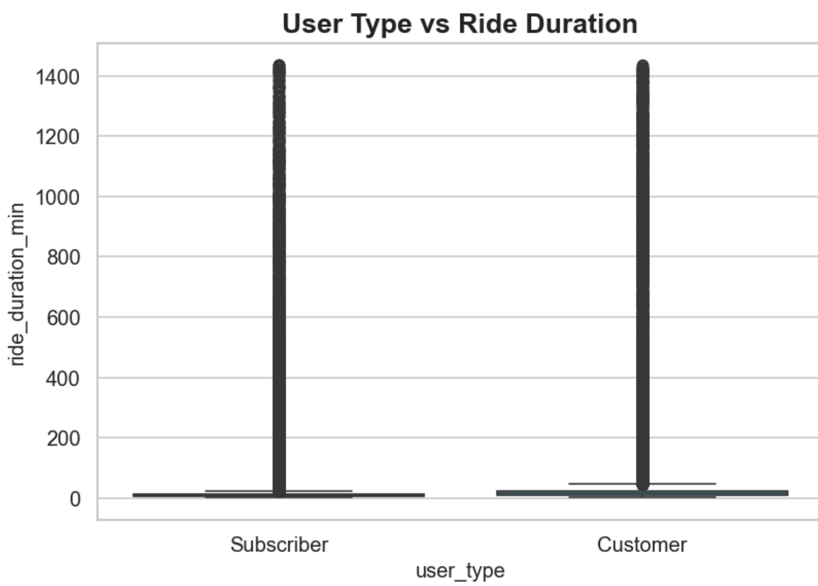

Figure 10.11: User Type vs Ride Duration

Metric	Observation
Subscriber Median Duration	Short, tightly clustered
Customer Median Duration	Slightly longer and more variable
Outliers	Present in both groups, extending to the maximum recorded duration

Table 10.11: Summary Statistics — User Type vs Ride Duration

Interpretation

This boxplot compares ride duration between Subscribers and Customers, and the difference between the two groups is immediately visible. Subscriber trips are tightly clustered around a low median duration with a narrow interquartile range, consistent with habitual, purposeful commuting behaviour where riders complete their journey efficiently and return the bike promptly. Customer trips, while still short on average, display a noticeably wider spread and a marginally higher typical duration, suggesting that casual users are more likely to take their time, potentially exploring the city or using the bike for leisure rather than a direct point-to-point commute. Both groups exhibit a considerable number of extreme outliers stretching toward the maximum possible ride duration, indicating that the lost-bike or extended-rental issue identified earlier is not confined to a single user category. This distinction between user types has practical pricing and product implications: it validates the rationale for offering habitual, short-trip-oriented pricing structures to Subscribers, while Customer-facing pricing or messaging could better acknowledge the somewhat longer, more exploratory nature of their typical usage.

Key Observations

- Subscribers show short, tightly clustered ride durations reflecting efficient commuting.
- Customers exhibit slightly longer and more variable ride durations.
- Extreme long-duration outliers appear in both user categories.
- The variation supports distinct behavioural profiles for each user type.

Business Recommendation

- Tailor pricing tiers to reflect the efficient, short-trip behaviour typical of Subscribers.
- Consider duration-aware pricing safeguards for Customers to manage extended-rental risk.

10.12 Gender vs Ride Duration

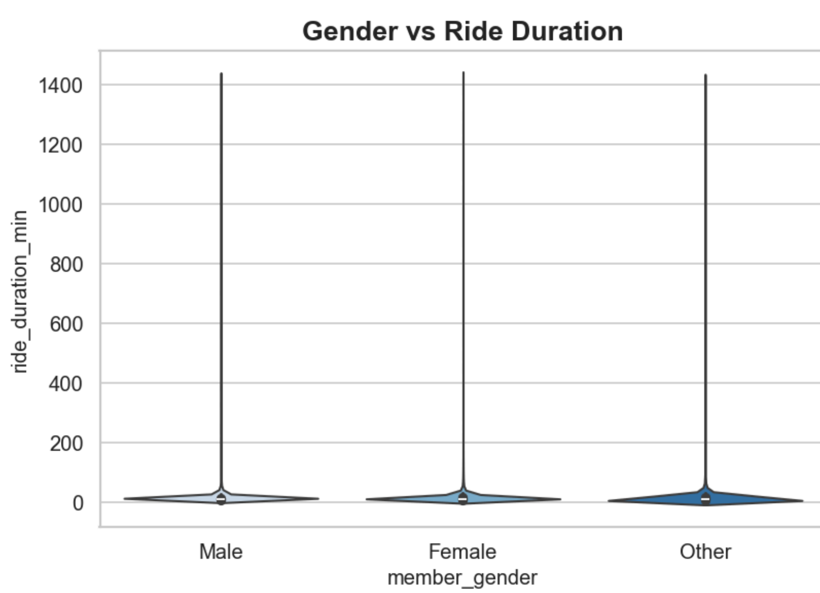

Figure 10.12: Gender vs Ride Duration

Metric	Observation
Male Riders	Similar distribution shape to overall population
Female Riders	Comparable median, marginally different spread
Other	Similar overall pattern despite smaller sample size

Table 10.12: Summary Statistics — Gender vs Ride Duration

Interpretation

This violin plot compares the distribution of ride duration across the three recorded gender categories. At a glance, the three distributions share a broadly similar shape: each is heavily concentrated at short durations near the bottom of the scale, with a narrow, elongated tail stretching upward to reflect the presence of outlier long-duration rides. The width and central tendency of the violins for Male and Female riders are closely comparable, indicating that typical ride length does not differ substantially by gender once the data has been cleaned and outliers accounted for. The Other category, while represented by a much smaller number of records, follows a broadly consistent pattern rather than diverging sharply from the other two groups. This similarity across gender categories suggests that ride duration is primarily driven by trip purpose and geography rather than by rider gender, which is a useful finding when designing gender-neutral service policies. It also means that the earlier observed gender imbalance in overall ridership is not accompanied by a meaningfully different usage pattern, reinforcing that outreach efforts aimed at underrepresented gender groups can focus on adoption and access rather than needing to accommodate a fundamentally different usage behaviour.

Key Observations

- Ride duration distributions are broadly similar across all three gender categories.
- Median duration does not differ substantially by gender.
- Long-duration outliers appear consistently across all groups.
- Gender therefore appears to have limited influence on typical trip length.

Business Recommendation

- Design service policies and pricing that remain gender-neutral, since usage behaviour does not differ meaningfully.
- Focus diversity outreach on adoption and access rather than on usage-pattern adjustments.

10.13 Age vs Ride Duration

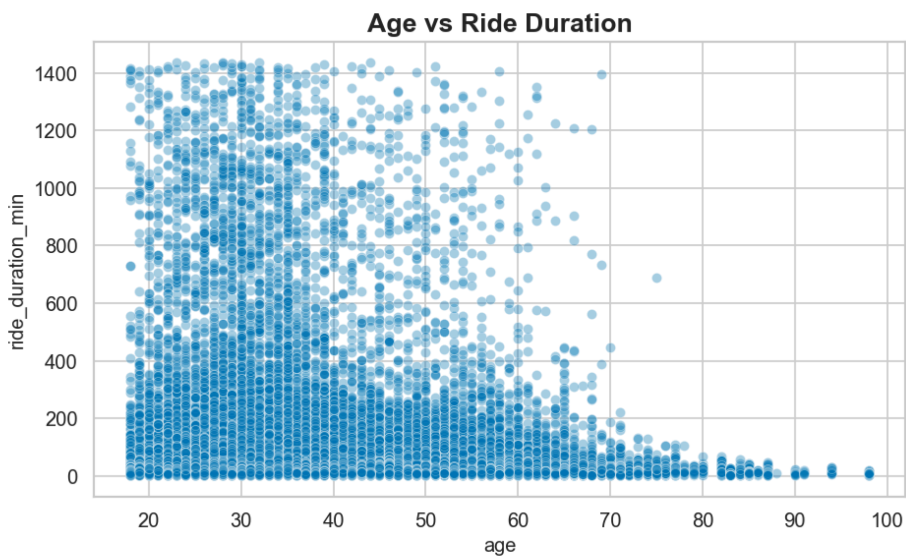

Figure 10.13: Age vs Ride Duration

Metric	Observation
Younger Riders (18–40)	Wide spread, including most long-duration outliers
Older Riders (60+)	Narrower spread, predominantly short durations
Overall Relationship	Weak, with no strong linear trend

Table 10.13: Summary Statistics — Age vs Ride Duration

Interpretation

This scatter plot examines the relationship between rider age and ride duration across the full dataset. The most visually striking feature is the concentration of long-duration outlier trips among younger and middle-aged riders, roughly between eighteen and fifty years old, with these extended rentals becoming progressively rarer as age increases. Riders beyond their sixties show a markedly tighter clustering around short durations, with very few extreme outliers appearing in this older segment. Despite this visual pattern in the outliers, the bulk of data points across all ages sit close to the lower end of the duration scale, suggesting that the core relationship between age and typical trip length is weak; most riders, regardless of age, complete short trips, and age appears to influence primarily the likelihood of an unusually long rental rather than typical day-to-day usage. This finding aligns with the correlation heatmap presented later in this report, which confirms only a negligible statistical association between these two variables. From a business perspective, this suggests that any operational policies addressing extended-duration rentals should not be narrowly targeted at a specific age group, since the core commuting behaviour is consistent across the age spectrum.

Key Observations

- Long-duration outlier trips are concentrated among younger and middle-aged riders.
- Riders aged 60 and above show a narrower, more consistent short-duration pattern.
- The overall relationship between age and duration is weak across the dataset.
- Typical short-trip behaviour is consistent across nearly all age groups.

Business Recommendation

- Avoid age-targeted policies for managing long-duration rentals; address the issue across the full rider base instead.
- Use this weak relationship to simplify pricing models, since age-based duration adjustments offer limited value.

10.3 Multivariate Analysis

10.14 Correlation Heatmap

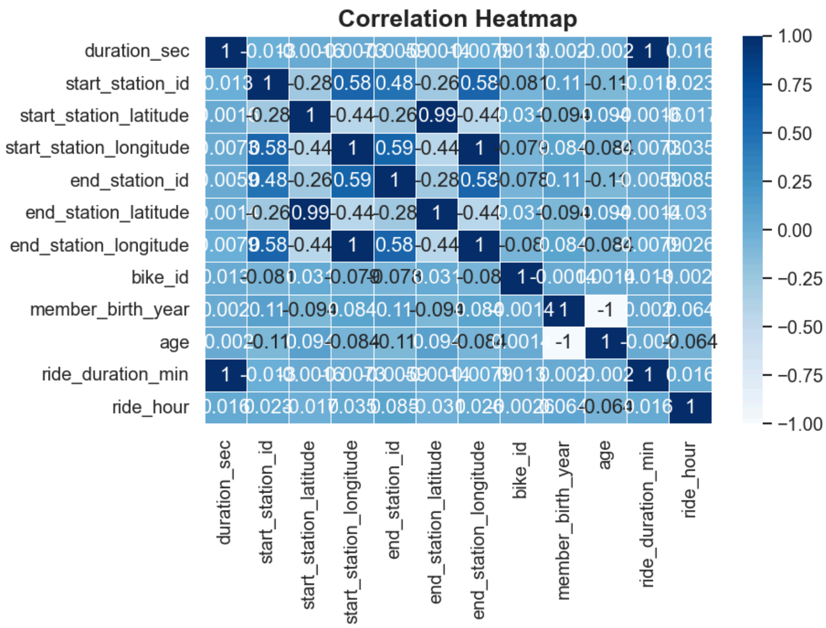

Figure 10.14: Correlation Heatmap

Metric	Observation
Strongest Positive Correlation	Station latitude and longitude pairs (~0.99)
Notable Negative Correlation	Member birth year and age (perfectly inverse, as expected)
Most Variables	Weak to negligible correlation with one another

Table 10.14: Summary Statistics — Correlation Heatmap

Interpretation

This heatmap presents pairwise correlation coefficients across all numerical variables in the cleaned dataset, offering a compact statistical overview of how strongly different attributes move together. As expected, station latitude and longitude values are very strongly correlated with themselves and with each other, since these coordinates are geographically linked and describe fixed physical locations. Similarly, member birth year and rider age display a perfect inverse relationship, which is a mathematical certainty given that age was directly derived from birth year rather than a genuine independent finding. Beyond these structurally expected relationships, the majority of variable pairs — including duration, ride hour, and station identifiers — show only weak to negligible correlation with one another, generally clustering close to zero. This is itself a meaningful finding: it indicates that ride duration is not strongly driven by any single other numerical factor captured in this dataset, reinforcing the earlier observation that trip length is more likely influenced by qualitative factors such as trip purpose or route choice rather than by station location, time of day, or rider age alone. For any future predictive modelling effort, this correlation structure suggests that simple linear relationships between these variables would have limited explanatory power, and more sophisticated feature engineering or non-linear modelling techniques may be required to accurately forecast ride duration or demand.

Key Observations

- Geographic coordinate pairs show very strong, expected positive correlation.
- Age and birth year show a perfect inverse relationship by construction.
- Most operational variables show weak or negligible correlation with one another.
- Ride duration is not strongly explained by any single numerical variable in isolation.

Business Recommendation

- Prioritise non-linear or feature-engineered modelling approaches for future demand or duration forecasting.
- Use the weak correlation structure to justify collecting richer contextual data, such as trip purpose or route.

10.15 Multivariate Relationship Pairplot

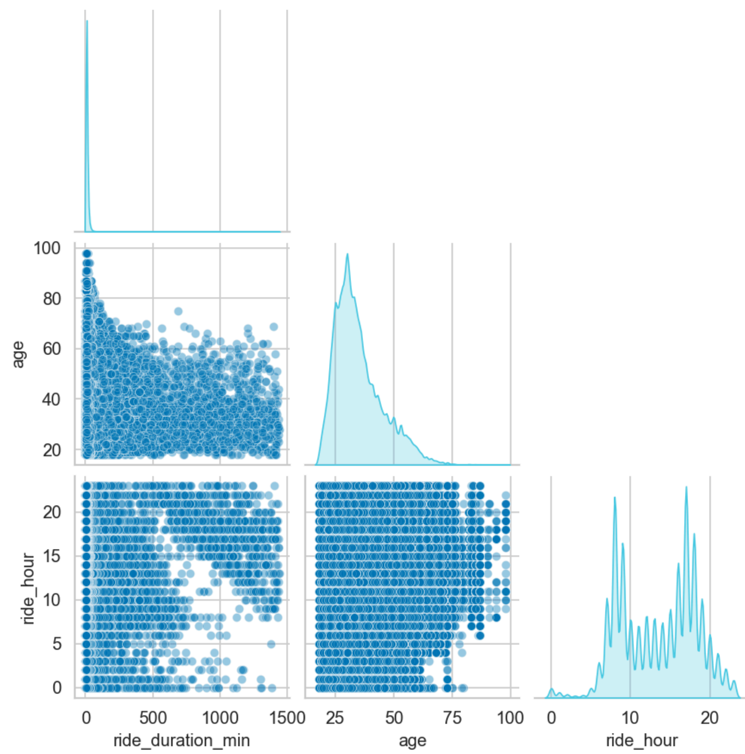

Figure 10.15: Multivariate Relationship Pairplot

Metric	Observation
Ride Duration	Sharply right-skewed, concentrated near zero
Age	Right-skewed, peaking in the late twenties to thirties
Ride Hour	Bimodal, with peaks aligning to commuting hours

Table 10.15: Summary Statistics — Multivariate Relationship Pairplot

Interpretation

This pairplot brings together ride duration, rider age, and ride hour in a single combined view, showing both the individual distribution of each variable along the diagonal and the pairwise scatter relationships between them. The diagonal panels confirm patterns already established elsewhere in this report: ride duration is sharply right-skewed and concentrated near very short values, age peaks in the late twenties to thirties before tapering off, and ride hour displays the same bimodal commuting pattern observed in the hourly ride distribution chart, with visible density around the morning and evening peaks. The scatter panels reinforce that no single pair of these variables exhibits a strong, easily describable linear relationship; points are broadly and evenly scattered across the age-versus-duration and hour-versus-duration panels, with the concentration of activity driven mainly by the marginal distributions rather than by any interaction effect between the variables. This combined view is valuable precisely because it consolidates several earlier findings into one visual summary, confirming that the dataset's key variables behave largely independently of one another and that Ford GoBike's ridership is fundamentally characterised by short, frequent trips taken by a working-age population during two clearly defined daily commuting windows.

Key Observations

- The pairplot confirms the sharply right-skewed nature of ride duration.
- Age distribution peaks among young working adults, consistent with earlier findings.
- Ride hour retains its clear bimodal, commuter-driven pattern in this combined view.
- No strong pairwise interaction effects are evident among these three variables.

Business Recommendation

- Treat duration, age, and ride hour as largely independent factors in future segmentation models.
- Use this consolidated view as a quick-reference diagnostic when onboarding new analysts to the dataset.

11. Business Insights

The exploratory analysis presented in the preceding section surfaces a coherent and consistent narrative about how the Ford GoBike service is actually used across the Bay Area. Subscribers dominate the platform, contributing the substantial majority of trips, which signals a stable, membership-anchored revenue base rather than one dependent on unpredictable casual usage. This is reinforced by the strongly weekday-oriented ridership pattern, with volumes peaking from Monday through Thursday and falling sharply over the weekend, and by the distinct bimodal hourly pattern that aligns tightly with typical morning and evening commuting windows. Together, these findings paint a picture of a service that functions primarily as a practical, everyday commuting tool rather than a leisure or tourism product.

The demographic analysis adds further texture to this picture. Riders are predominantly young to middle-aged adults, with the heaviest concentration falling between the mid-twenties and mid-thirties, and the rider base skews considerably male, at roughly three-quarters of all identified users. Ride duration, meanwhile, is overwhelmingly short, with the vast majority of trips completed in well under fifteen minutes, again consistent with point-to-point commuting rather than extended recreational use. A notable secondary finding is the presence of a persistent, if numerically small, population of extreme-duration outlier trips, some approaching a full twenty-four hours, which likely reflect unreturned or lost bikes rather than genuine continuous rides and merit dedicated operational attention.

Station-level analysis reveals that demand is concentrated around a relatively small set of high-traffic hubs, most of which sit near major transit interchanges such as Caltrain stations, the Ferry Building, and BART connections. These stations serve simultaneously as popular origins and destinations, indicating that Ford GoBike functions as a critical first-and-last-mile connector within the broader Bay Area transit ecosystem rather than as a standalone mode of transportation. Finally, the correlation and multivariate analysis confirms that most numerical variables in the dataset are only weakly related to one another, meaning that ride duration and demand are shaped by a combination of contextual factors — trip purpose, route, and individual habit — that are not fully captured by the structured fields alone, pointing toward valuable directions for future data collection and modelling.

Finding	Business Impact
Subscribers dominate total ridership	Provides a stable, recurring revenue foundation
Ride demand peaks during weekday commuting hours	Informs fleet sizing and staff scheduling decisions
Ridership skews toward younger working adults	Sharpens the target audience for marketing campaigns
A small set of transit-hub stations drives most activity	Guides prioritised bike redistribution and dock capacity

Table 11.1: Key Findings and Associated Business Impact

12. Business Recommendations

Based on the findings obtained from this Exploratory Data Analysis, the following recommendations are proposed to help Ford GoBike improve operational efficiency, strengthen customer relationships, and support long-term strategic planning. These recommendations are organised around the three central themes that emerged from the analysis: fleet and station management, customer growth and retention, and forward-looking analytical capability.

On the operational side, since ride demand is heavily concentrated during predictable weekday commuting windows and around a well-defined set of transit-adjacent stations, Ford GoBike should implement a proactive bike redistribution strategy that repositions bikes toward high-demand stations ahead of the morning and evening peaks. This targeted approach is likely to be considerably more cost-effective than uniform, system-wide redistribution, since it concentrates limited staffing and logistics resources where they will have the greatest impact on availability and customer satisfaction. Complementary to this, a dedicated monitoring process should be established to flag rentals that exceed a reasonable duration threshold, allowing operations staff to intervene early on potential lost-bike incidents rather than discovering them only through the extreme-outlier trips identified in this analysis.

From a growth perspective, the overwhelming reliance on Subscribers represents both a strength to protect and an opportunity to expand. Referral incentives, loyalty rewards, and clearly communicated cost savings relative to casual pay-per-ride pricing could help convert more of the existing Customer base into long-term subscribers. At the same time, the pronounced gender imbalance in ridership suggests a meaningful, currently underserved opportunity: targeted safety-focused messaging, women-oriented promotional campaigns, and continued investment in well-lit, secure station environments could help broaden the platform's appeal. Finally, given the relatively weak correlations observed among the current set of variables, Ford GoBike should invest in expanding its data collection to include contextual attributes such as trip purpose, weather conditions, or connecting transit usage, which would materially strengthen any future predictive modelling or demand-forecasting initiative built on top of this analytical foundation.

Recommendation	Benefit
Targeted Bike Redistribution	Improved bike availability at high-demand stations
Subscriber Conversion Promotions	Accelerated customer growth and revenue stability
Predictive Analytics Investment	More accurate demand forecasting and planning
Extended-Duration Rental Monitoring	Reduced lost-bike incidents and billing anomalies

Table 12.1: Recommendations and Expected Benefits

13. Challenges Faced

Several practical challenges were encountered over the course of this project, each requiring careful judgment to resolve without compromising the integrity of the analysis. The first challenge arose during data integration, where twelve separately formatted monthly files needed to be reliably merged into a single dataset; ensuring consistent column naming and data types across all twelve files required attentive validation before the concatenation step could be trusted. A second, more substantial challenge emerged during data cleaning, where a meaningful proportion of records were missing gender or birth-year information. Because these fields underpin much of the demographic analysis presented in this report, a decision had to be made between imputing missing values, which risked introducing bias, and removing incomplete records, which reduces the effective sample size; removal was ultimately chosen as the more conservative and defensible approach.

A further challenge was posed by the presence of extreme outlier ride durations, some approaching a full twenty-four hours. These values are almost certainly not representative of genuine cycling activity, yet arbitrarily removing them risked discarding potentially useful information about lost or delayed bike returns. Rather than deleting these records outright, they were retained and explicitly analysed as a distinct pattern of interest within the duration-focused visualisations. Finally, working with a dataset exceeding two million rows introduced practical performance considerations during visualisation, particularly for detailed scatter and pair plots, requiring careful attention to rendering settings to keep the analysis both accurate and computationally manageable within a standard notebook environment.

14. Future Scope

While this project provides a comprehensive exploratory foundation, several avenues remain open for extending the analysis into more advanced and operationally embedded capabilities. A natural next step is the application of machine learning techniques to build predictive models for ride demand, allowing Ford GoBike to anticipate station-level bike requirements hours or even days in advance rather than relying solely on historical averages. Closely related to this is the opportunity to develop dedicated demand prediction models that incorporate external factors such as weather forecasts, local events, and public transit schedules, which were identified in this analysis as likely contributors to ride patterns not yet captured within the existing dataset.

Beyond predictive modelling, there is significant value in operationalising these insights through an interactive Power BI dashboard, enabling operations and strategy teams to monitor key metrics such as station utilisation, subscriber growth, and duration outliers in real time rather than through periodic static reports. Complementary interactive maps could visualise station-level demand and rebalancing needs geographically, making station prioritisation more intuitive for field teams. Further, applying customer segmentation techniques, such as clustering algorithms, could reveal more nuanced rider personas beyond the broad Subscriber and Customer categories used in this report, supporting more precisely targeted marketing. Finally, combining AI-based forecasting with the demand and segmentation work described above would position Ford GoBike to move from a reactive, historically informed operating model to a proactive, forward-looking one capable of anticipating and shaping demand rather than simply responding to it.

15. Conclusion

This project successfully carried out a comprehensive Exploratory Data Analysis of one full year of Ford GoBike trip data, progressing methodically from raw, fragmented monthly files through integration, cleaning, feature engineering, and finally a rich set of univariate, bivariate, and multivariate visualisations. Each stage of the pipeline was designed to build a progressively clearer and more trustworthy picture of how the bike-share service is actually used across the

Bay Area, ensuring that the insights ultimately presented rest on a solid analytical foundation rather than on raw, unexamined data.

The findings that emerged were both consistent and actionable. Subscribers were shown to anchor the platform's ridership, contributing the vast majority of trips and pointing to a stable, membership-driven revenue base. Ride activity was found to be strongly concentrated during weekday commuting hours and around a defined set of transit-adjacent stations, confirming the service's role as a practical first-and-last-mile connector rather than a leisure product. Ride durations were overwhelmingly short, punctuated by a smaller population of extreme-duration outliers warranting operational follow-up, and demographic patterns revealed a young, working-age, and predominantly male user base offering clear direction for future marketing and inclusion efforts.

Taken together, these findings translate directly into a set of concrete recommendations spanning bike redistribution, subscriber growth, outlier monitoring, and expanded data collection for future predictive work. More broadly, this project demonstrates how a well-executed exploratory analysis can convert a large, raw operational dataset into a genuinely useful decision-support resource, illustrating the practical value of data analytics within the shared-mobility sector and providing a strong foundation for the more advanced modelling and dashboard initiatives outlined in the future scope of this report.

16. References

- Ford GoBike Trip Data — Publicly released monthly trip datasets used as the basis for this analysis.
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